LAB 4

Objective:

1. Write an assembly language program in 8085 microprocessors to find square of 8-bit number.
2. Write an assembly language program to find 1's complement of a number.
3. Write an assembly language program to find 2's complement of a number.
4. The data 20H and 30H are stored in 2050H and 2051H. WAP to transfer the data to 3000H and 3001H using LHLD and SHLD instructions.
5. Pair B contains 1122H and pair D contains 3344H. WAP to exchange the contents of B and D pair using XCHG instruction.
6. WAP to add two 4 digit BCD numbers equals 7342 and 1989 and store result in BC register.

Write an assembly language program in 8085 microprocessors to find square of 8-bit number.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	26	MVI H, C0	Initialize register H with C0
C001	C0		
C002	2E	MVI L, 50	Initialize register L with 50
C003	50		
C004	3E	MVI A, 00	Initialize accumulator A with 00
C005	00		
C006	46	MOV B, M	Moves the content of memory location which is indirectly specified by M in register B
C007	86	X: ADD M	Add the content of memory location which is indirectly specified by M in accumulator A
C008	05	DCR B	Decrement value of register B by 1
C009	C2	JNZ X	Jump to memory location 2007 if ZF = 0, i.e. register B does not contain 0
C00A	07		
C00B	00		
C00C	32	STA C055	Stores the content of accumulator in C055
C00D	55		
C00E	C0		
C00F	76	HLT	Terminate the program.

Input:

Memory Location Data

C050H 04H

Output:

Memory Location Data

C055H 10H

Write an assembly language program to find 1's complement of a number.

Statement: Input number from memory location C013H and store result in memory location C052H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3A	LDA C013	Load contents from memory location C013H to accumulator
C001	13		
C002	C0		
C003	2F	CMA	Complement contents of accumulator
C004	32	STA C052	Store result in memory location C052H
C005	52		
C006	C0		
C007	76	HLT	Terminate the program.

Input:

Memory Location Data

C013H 12H

Output:

Memory Location Data

C052H ED H

Write an assembly language program to find 2's complement of a number.

Statement: Input number from memory location C013H and store result in memory location C052H.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3A	LDA C013	Load contents from memory location C013H to accumulator
C001	13		
C002	C0		
C003	2F	CMA	Complement contents of accumulator
C004	C6	ADI 01	Add 01H to the contents of accumulator
C005	01		
C006	32	STA C052	Store result in memory location C052H
C007	52		
C008	C0		
C009	76	HLT	Terminate the program.

Input:

Memory Location Data

C013H 12 H

Output:

Memory Location Data

C052H EE H

The data 20H and 30H are stored in 2050H and 2051H. WAP to transfer the data to 3000H and 3001H using LHLD and SHLD instructions.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	3E	MVI A, 20	Load the immediate data 20 to accumulator
C001	20		
C002	32	STA 2050	Store the content of accumulator to 2050
C003	50		
C004	20		
C005	3E	MVI A,30	Load the immediate data 30 to accumulator
C006	30		
C007	32	STA 2051	Store the content of accumulator to 2051
C008	51		
C009	20		
C00A	2A	LHLD 2050	This instruction copies the contents of memory location pointed out by 16-bit address 2050 into register L and copies the contents of next memory location 2051 into register H.
C00B	50		
C00C	20		
C00D	22	SHLD 3000	The contents of register L are stored into memory location specified by the 16-bit address 3000. The contents of register H are stored into the next memory location 3001.
C00E	00		
C00F	30		
C010	76	HLT	Terminate the program.

Input:

Memory Location Data

2050 20

2051 30

Output:

Memory Location Data

3000 20

3001 30

Pair B contains 1122H and pair D contains 3344H. WAP to exchange the contents of B and D pair using XCHG instruction.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	01	LXI B, 1122	Load 16-bit data 1122H to BC pair
C001	22		

C002	11		
C003	11	LXI D, 3344	Load 16-bit data 3344H to DE pair
C004	44		
C005	33		
C006	60	MOV H, B	Move the content of register B to H
C007	69	MOV L, C	Move the content of register C to L
C008	EB	XCHG	Exchange DE pair with HL pair
C009	44	MOV B,H	Move the content of register H to B
C00A	4D	MOV C, L	Move the content of register L to C
C00B	76	HLT	Terminate the program.

Input:

Content of Register Before Execution

B =11, C=22, D = 33, E =44

Output:

Content of Register After Execution

B= 33, C= 44, D= 11, E= 33, H=33, L= 44

WAP to add two 4 digit BCD numbers equals 7342 and 1989 and store result in BC register.

Address	Hex Code (Opcode)	Mnemonics	Comments
C000	21	LXI H, 7342	Load 16-bit data 7342H to HL pair
C001	42		
C002	73		
C003	01	LXI B, 1989	Load 16-bit data 1989H to BC pair
C004	89		
C005	19		
C006	7D	MOV A, L	Move the content of L to accumulator
C007	81	ADD C	Add the content of C with Accumulator
C008	27	DAA	Decimal adjustment accumulator: adjust after addition
C009	4F	MOV C, A	Move the content of accumulator to register C
C00A	7C	MOV A, H	Move the content of register H to accumulator
C00B	88	ADC B	Add the content of register B with carry to accumulator
C00C	27	DAA	Decimal adjustment accumulator: adjust after addition
C00D	47	MOV B, A	Move the content of accumulator to register B
C00E	76	HLT	

Input:

Content of Register Before Execution

B =19, C=89, H = 73, L =42

Output:

Content of Register After Execution

B= 93, C= 31